

BIOS 640: Assignment Week 3

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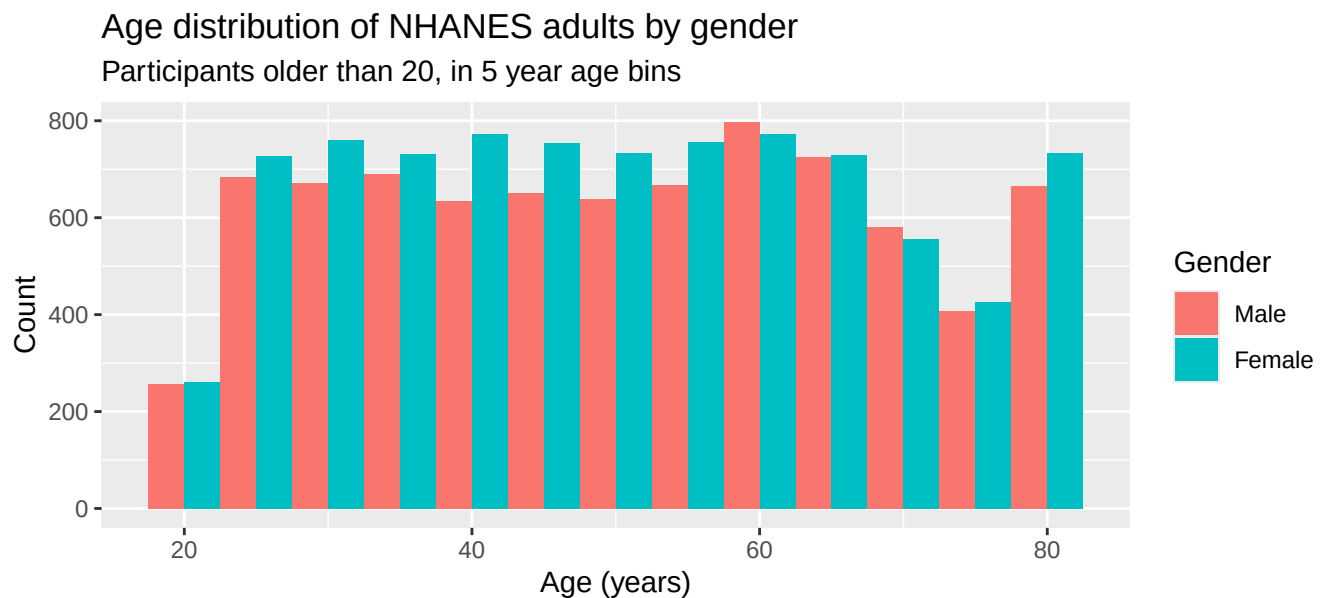
October 03, 2026

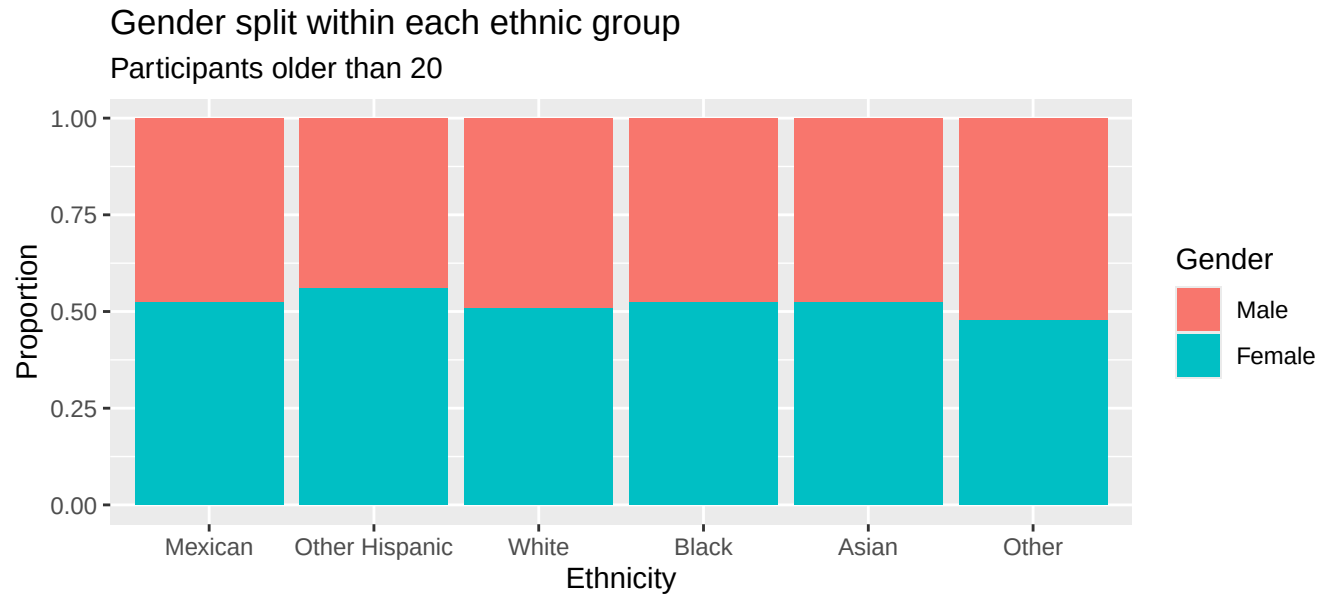
Exercise 1: Reproducing and arranging ggplot2 figures

Recreating the two figures

The two target figures only show adults. The first age bin (17.5 to 22.5 years) holds about 260 people per gender, which matches the data exactly when only participants older than 20 are kept, so ages 21 and 22 fall in that bin. I therefore restrict both figures to participants aged over 20.

The raw NHANES codes are replaced by the cleaned labels. Gender code 1 is "Male" and code 2 is "Female", so I set that factor order to keep the original colours. Ethnicity is ordered by the original RIDRETH3 codes (1 Mexican, 2 Other Hispanic, 3 White, 4 Black, 6 Asian, 7 Other).





Combining the figures

Comparing `plot_grid()` and `ggarrange()`

Both functions stack the plots in a single column and add panel labels. Each needs its own way to add an overall title, a `draw_label()` row for `plot_grid()` and `annotate_figure()` for `ggarrange()`. With `align = "v"` they also line up the plotting areas, so the two x axes start at the same position even though the y axis labels have different widths.

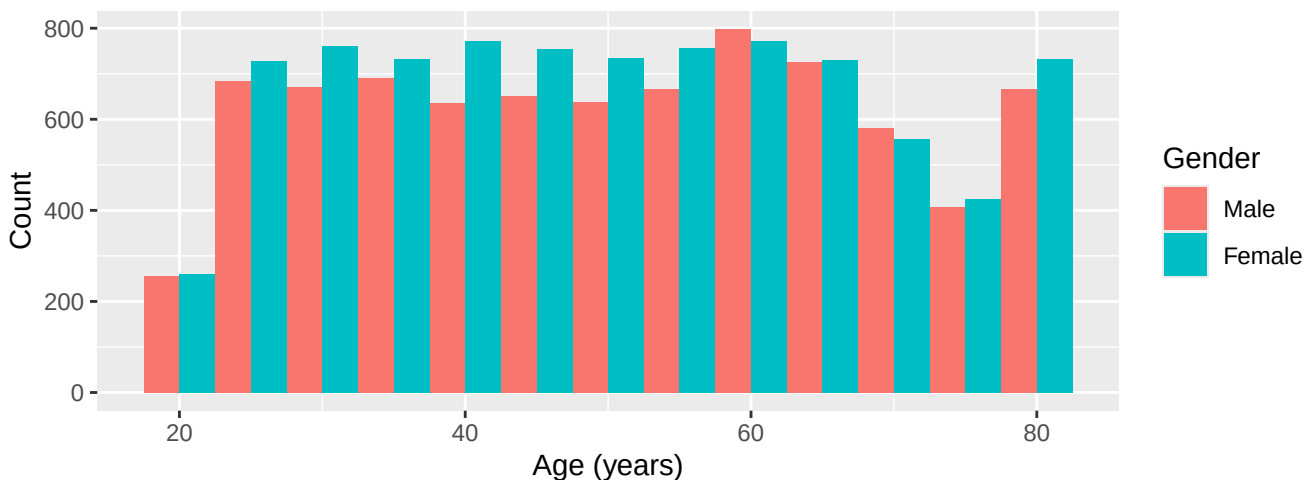
The main difference is how the legends are handled. `plot_grid()` keeps a separate gender legend beside each panel, so the same key appears twice and uses space that could go to the data. `ggarrange()` offers `common.legend = TRUE`, which draws one shared legend and gives both panels the same width.

For these two plots `ggarrange()` gives the better layout. Both panels map gender to the same colours, so a single legend is enough and the figure reads as one unit. `plot_grid()` would be my choice when the panels have different legends or when I need finer control over relative widths and heights.

Age and ethnicity of NHANES adults by gender

A Age distribution of NHANES adults by gender

Participants older than 20, in 5 year age bins



B Gender split within each ethnic group

Participants older than 20

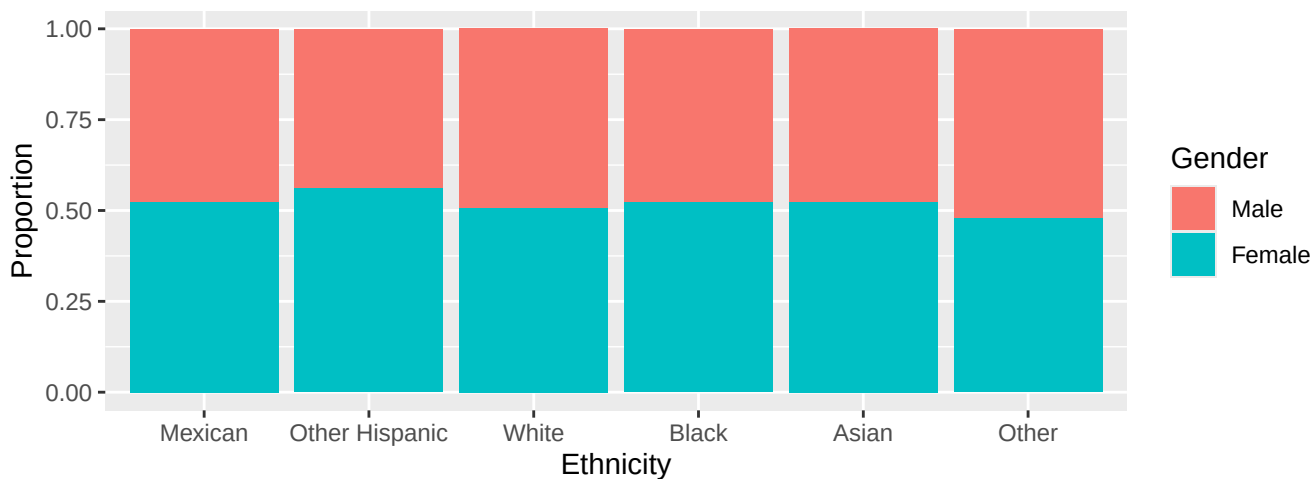


Figure 1: Figures combined with `plot_grid()` from `cowplot`.

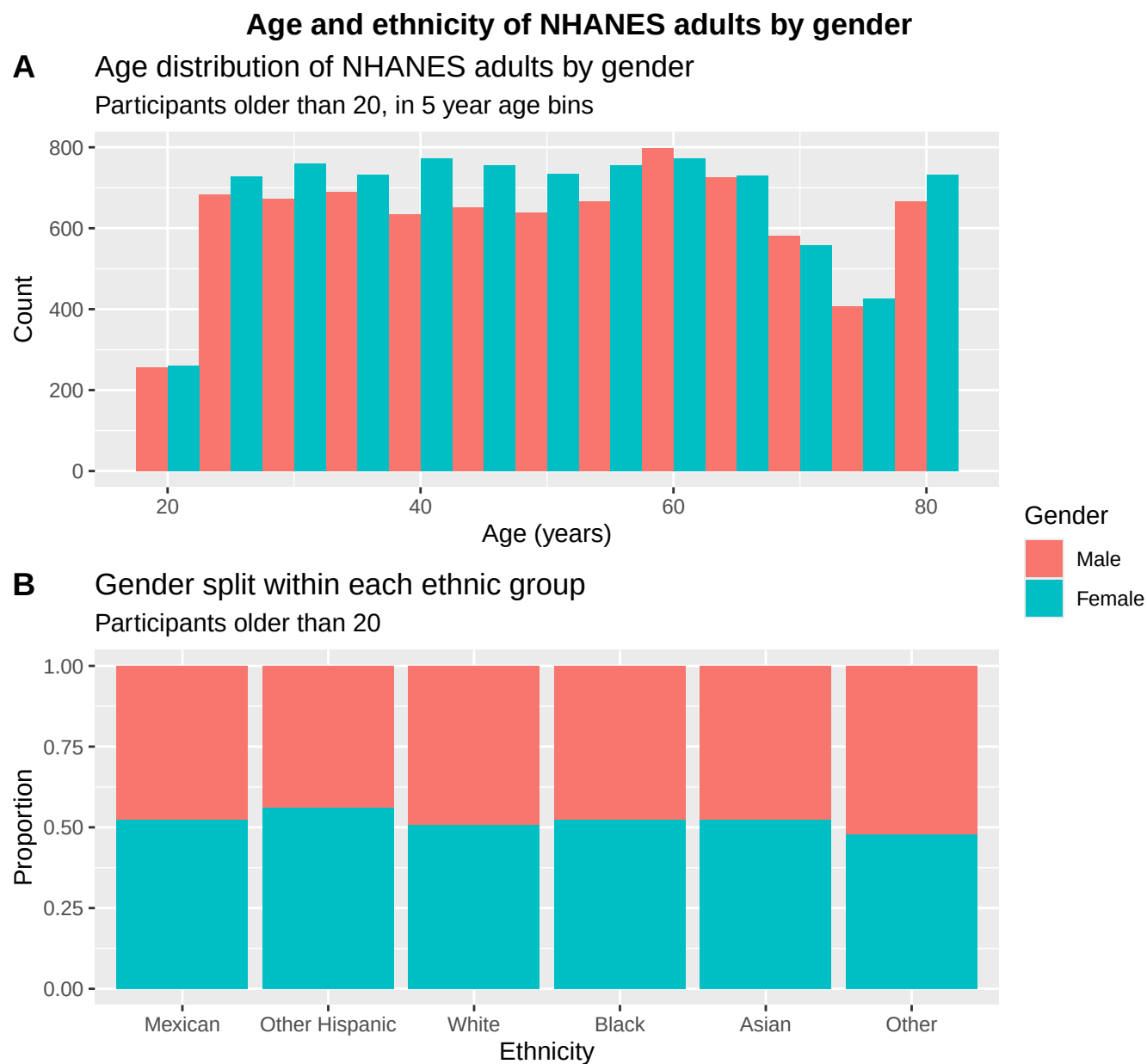


Figure 2: Figures combined with `ggarrange()` from `ggpubr`.

Exercise 2: Visualizing key characteristics

Table 1: Number of missing values per variable.

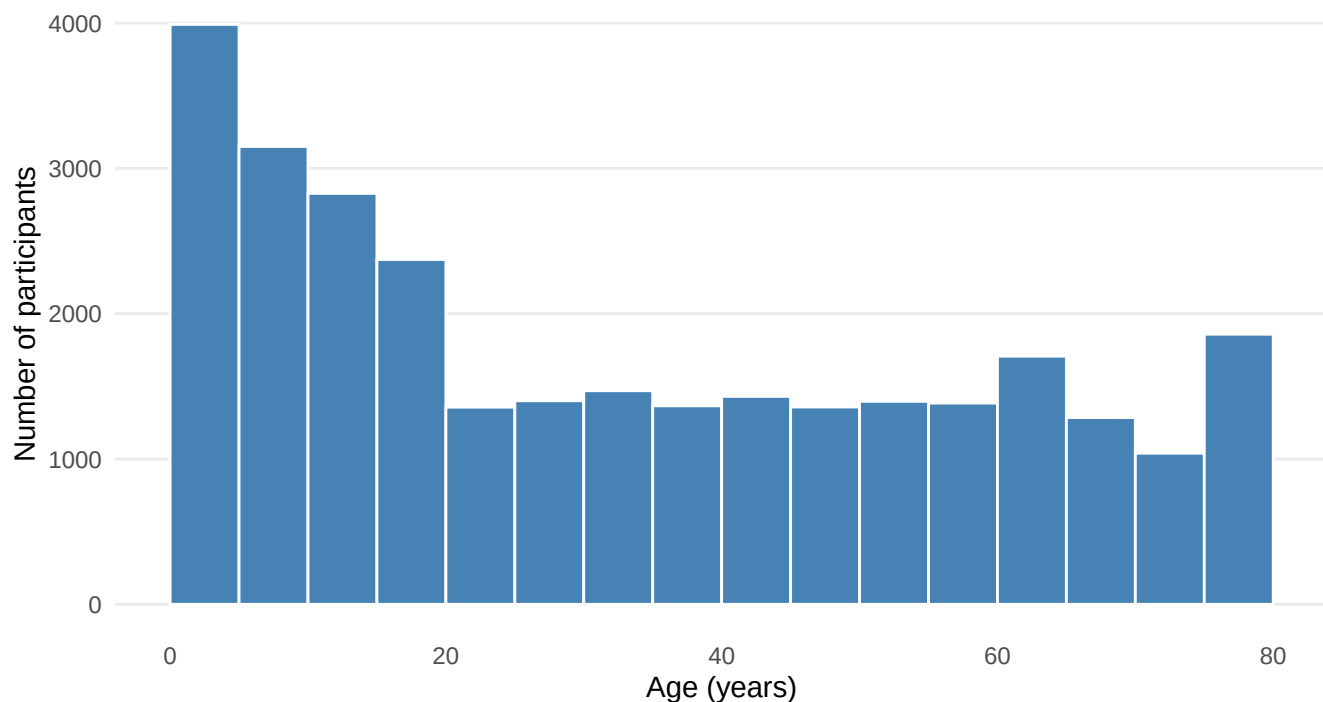
age	gender	ethnicity_1	ethnicity_2
0	0	0	0

None of the four variables has missing values, so all 29,400 participants contribute to each plot below.

Age

Age distribution of NHANES participants

Children make up a large share, and ages 80 and over are pooled at 80



Source: NHANES, all participants

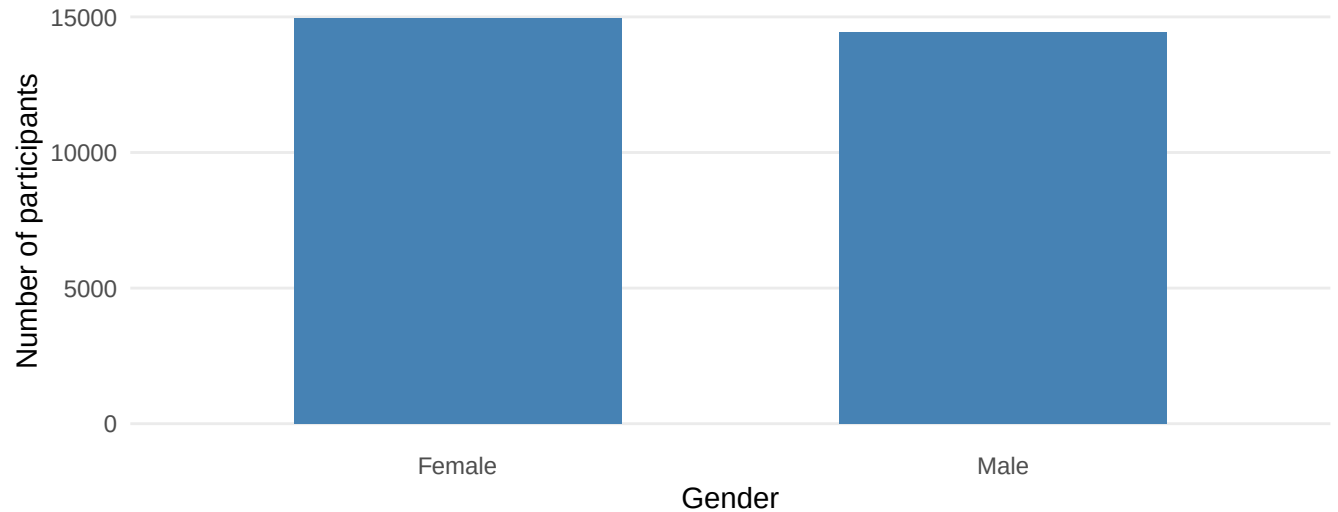
Age is continuous, so a histogram with 5 year bins is appropriate. The distribution is right skewed, with a median of 28 years and a mean of 32.5 years. Children and teenagers make up a large share of the sample (38.9% are under 18).

Counts fall steadily after about age 60, but the last bin rises again. NHANES top codes age at 80, so everyone aged 80 or older is recorded as 80 and piles into that final bin.

Gender

Number of NHANES participants by gender

Females and males are almost equally represented



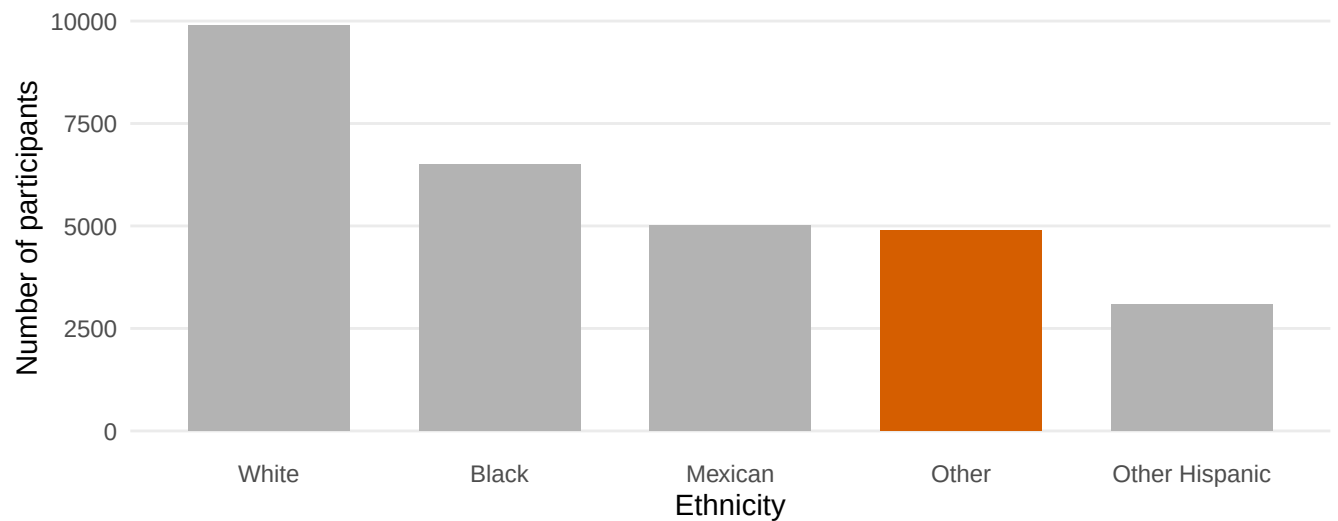
Source: NHANES, all participants

Gender is categorical, so a bar chart of counts is used. The sample is nearly balanced, with 14948 females (50.8%) and 14452 males.

Ethnicity

Participants by ethnicity (ethnicity_1)

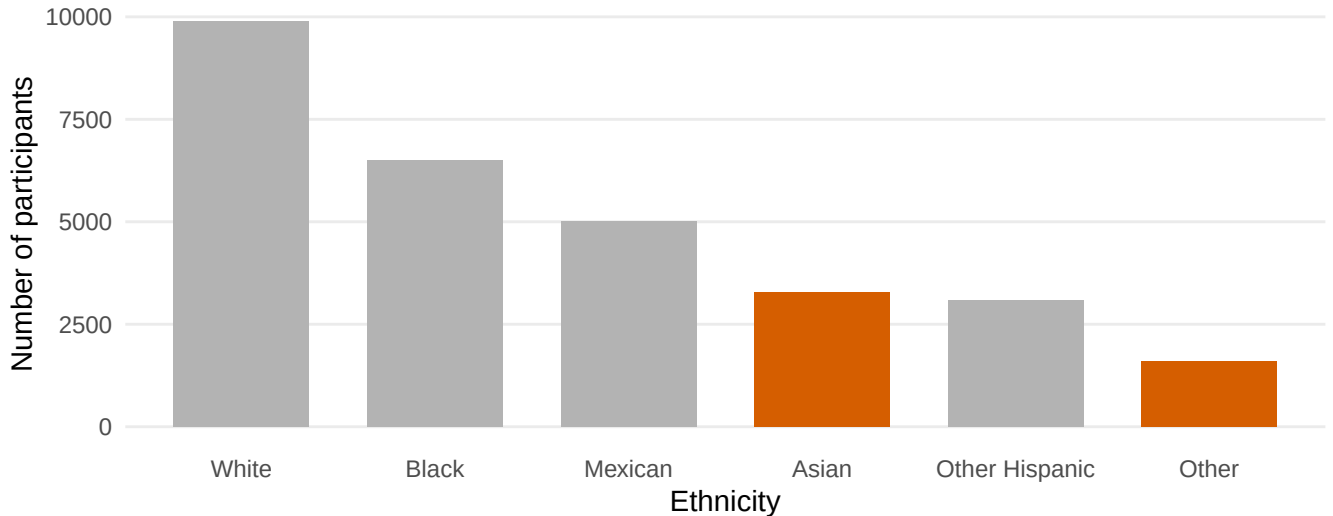
The Other group mixes Asian participants with everyone else



Source: NHANES, all participants

Participants by ethnicity (ethnicity_2)

Asian participants get their own category



Source: NHANES, all participants

Both ethnicity variables are categorical, so bar charts sorted from most to least frequent are used. Bars shared by both variables are grey, and the orange accent marks the categories where the two variables differ, so the comparison below is visible at a glance.

In `ethnicity_1`, White is the largest group (9890 participants), followed by Black and Mexican. The `ethnicity_2` variable has the same counts for those groups but has six categories instead of five.

Choosing an ethnicity variable

Table 2: Cross tabulation of the two ethnicity variables.

	Asian	Black	Mexican	Other	Other Hispanic	White
Black	0	6511	0	0	0	0
Mexican	0	0	5018	0	0	0
Other	3284	0	0	1609	0	0
Other Hispanic	0	0	0	0	3088	0
White	0	0	0	0	0	9890

The cross tabulation shows that the two variables agree for every participant except those coded “Other” in `ethnicity_1`. That group of 4893 people is split in `ethnicity_2` into 3284 Asian participants and 1609 participants in a smaller “Other” group.

I keep `ethnicity_2`. It holds all the information in `ethnicity_1` plus a separate Asian category, which is the fifth largest group in the sample and would otherwise be hidden inside a mixed “Other” group. A catch all category that large is hard to interpret, and health outcomes such as blood pressure can differ between Asian participants and the remaining groups.

Exercise 3: Improving a ggplot figure

What needs to improve

The research question is how each participant's weight changed relative to their own baseline. The colleague's figure plots absolute weight, so every line starts at a different height between about 64 and 100 kg. The reader has to mentally subtract each starting value to judge change, and between person differences in baseline weight dominate the picture.

The figure uses one colour per participant, which gives a legend of about twenty similar hues. Neighbouring shades of green, blue and pink are impossible to match back to their lines, so the legend adds clutter without helping identification. The legend also appears to be cut off, since participant 1 is not listed.

Several participants overlap. Five participants have perfectly flat lines, and two of them start near 78 kg, so their lines sit on top of each other and look like one thicker line. Without a reference line there is also no quick way to see who gained and who lost weight.

The labelling is minimal. There is no title, the y axis says "Weight" without units, and the x axis only marks weeks 0, 5, 10 and 15 even though the study ran for 16 weeks. The grey default background also lowers the contrast of the thin coloured lines.

Planned changes and why

I plot the change from baseline (weight at week t minus weight at week 0) in kilograms. Every line then starts at zero, which directly answers the question and removes the baseline differences. A dashed reference line at zero separates weight loss from weight gain at a glance.

Instead of twenty colours, I colour each line by its outcome at week 16: weight lost, no change or weight gained. Blue and orange from the colour blind friendly Okabe-Ito palette mark the two groups that changed, while grey keeps the unchanged group in the background. These colours are easy to tell apart, and the group label is printed directly next to the lines so no legend lookup is needed. A thick line shows the mean trajectory of each group while thin, semi transparent lines keep individual trajectories visible.

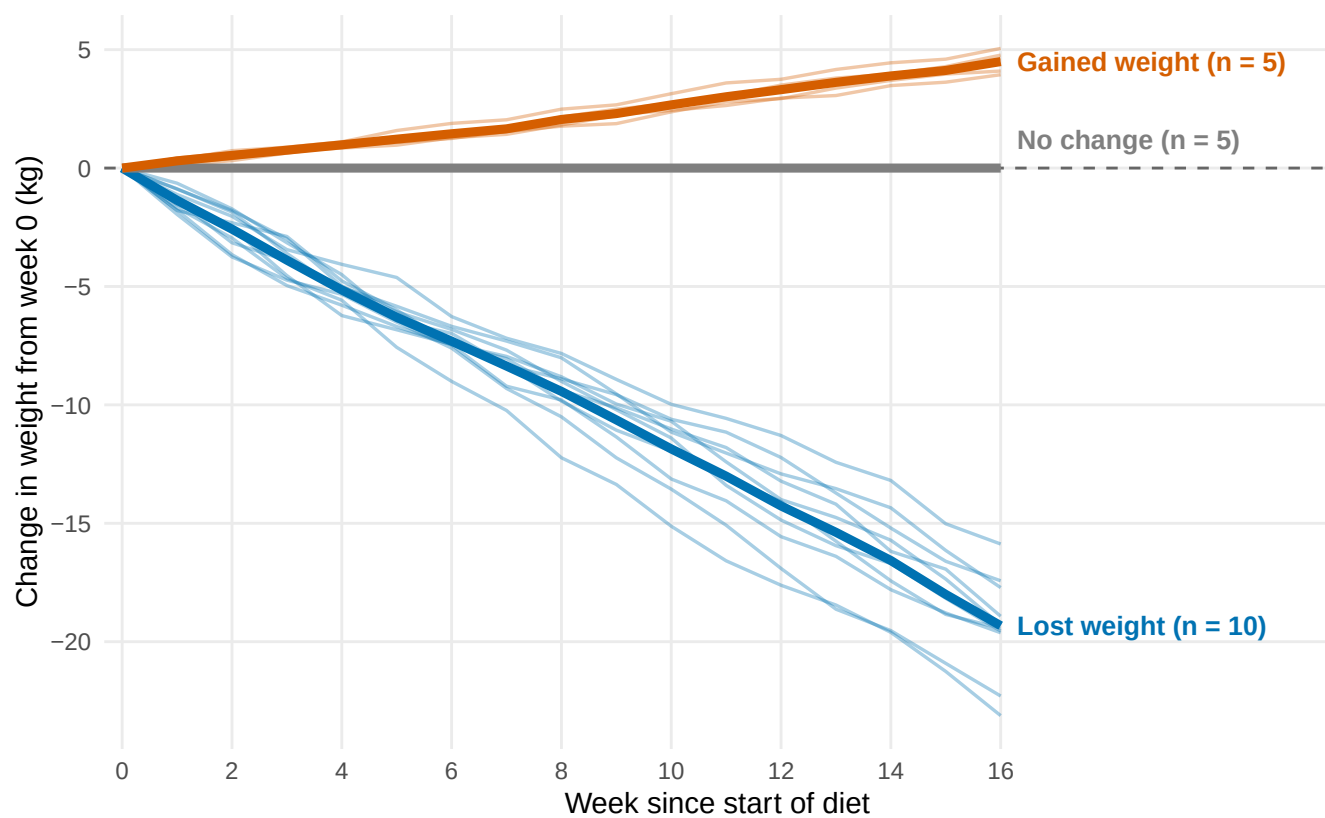
Finally, I add a descriptive title and subtitle, units on the y axis, x axis breaks every two weeks up to week 16, and a clean white theme. These choices make the figure self explanatory when read on its own.

Improved figure

The redesigned figure shows that 10 of the 20 participants lost weight, with an average change of -19.3 kg by week 16. The remaining participants either stayed exactly at their baseline weight or gained a few kilograms, which the colleague's figure hid among the overlapping lines.

Most participants lost weight over the 16 week diet

Thin lines show individual participants; thick lines show the group mean



Source: diet study, 20 participants

Figure 3: Change in body weight from baseline for each participant over the 16 week diet.

Exercise 4: Age, gender and systolic blood pressure

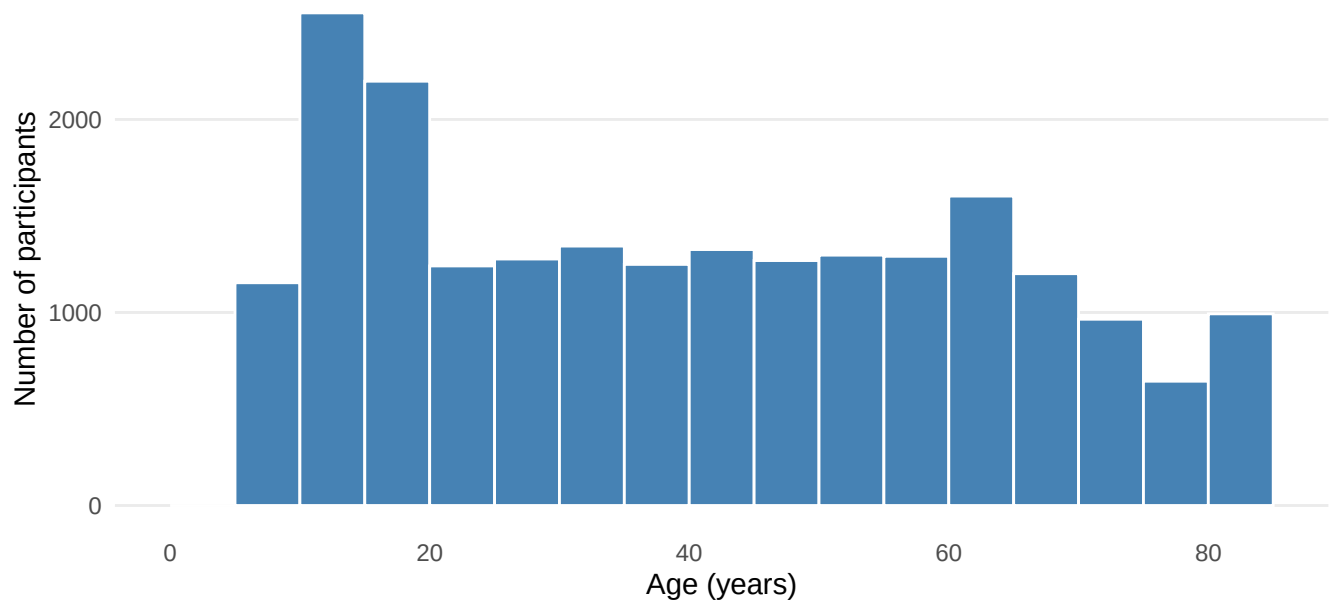
Who has a blood pressure measurement?

Systolic blood pressure (SBP) was averaged over up to four readings per participant, and a person counts as missing only when all four readings are absent. Of the 29,400 NHANES participants, 7,796 had no SBP reading at all, which leaves an analytic sample of 21,604 people (73.5%). No one was lost because of missing age or gender.

The missing readings are far from random with respect to age. NHANES only measures blood pressure from age 8, so the youngest participant in the final sample is 8 years old.

Age distribution in the final sample

No readings below age 8



Source: NHANES, participants with average SBP

Figure 4: Age distribution in the final blood pressure sample, redrawn from the Exercise 2 age histogram.

The large spike of young children in the full sample disappears, and the median age rises from 28 to 39 years. From age 10 upward about nine in ten participants have a reading, so above that age the two histograms keep the same shape and the loss is concentrated in early childhood. Any statement about blood pressure in children therefore rests on older children and teenagers only.

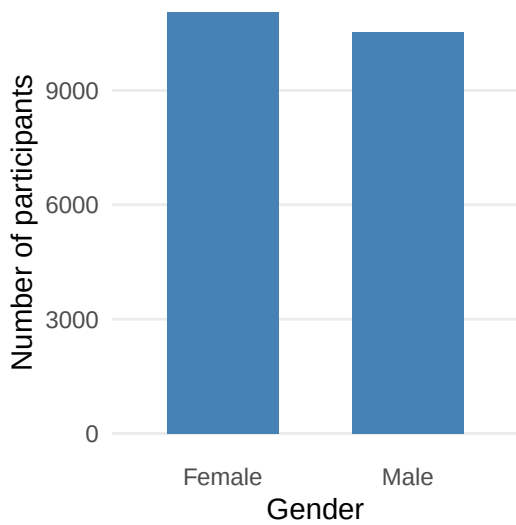
The gender balance barely moves, with 51.2% females in the final sample compared with 50.8% before filtering. Most participants have a normal SBP (56.9%). Still, about one person in four falls into one of the two hypertension stages, and Stage 2 is more common than Stage 1.

How blood pressure changes with age and gender

In both genders the pale band for normal SBP shrinks with every older age group, while the dark red Stage 2 band grows. Among people aged 60 and over, only 21% of women and 24% of men still have a normal reading.

Participants by gender

Still close to an even split



Participants by SBP category

About one in four has hypertension

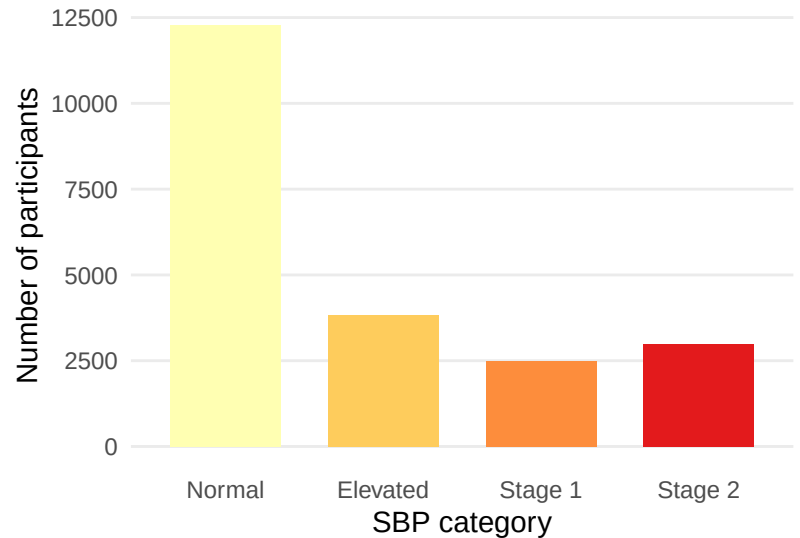


Figure 5: Gender balance and blood pressure categories in the final sample.

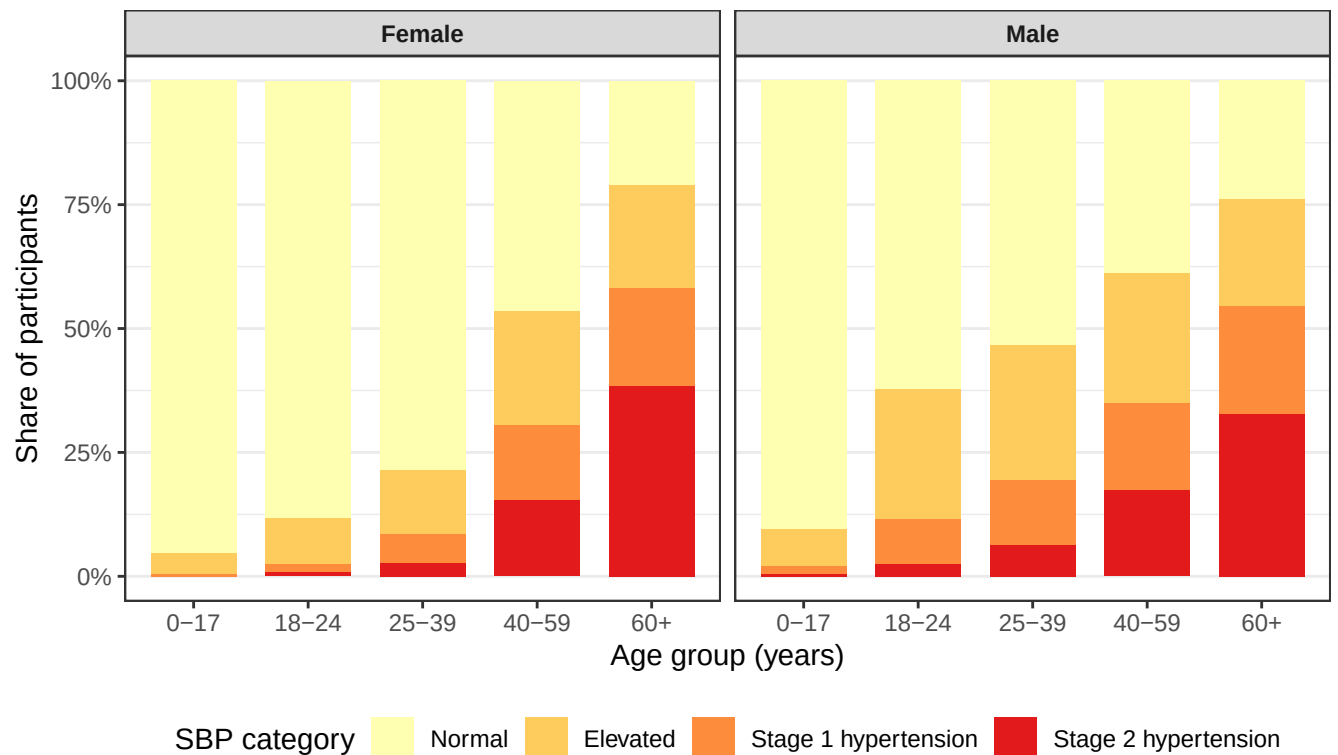
The two genders follow different paths to that point. Men move out of the normal range much earlier, and already 38% of men aged 18 to 24 have at least elevated SBP compared with 12% of women the same age.

Women stay mostly normal until their forties, then rise quickly, a sharp upward trajectory often attributed to the vascular and metabolic shifts of perimenopause and menopause, such as the loss of estrogen's natural protective effects on blood vessel elasticity. Consequently, by 60 and over, Stage 2 hypertension is more common in women (38%) than in men (33%).

Taken together, age is the strongest driver of high SBP in this sample, and gender changes the timing of that rise. These are unweighted descriptive patterns, and adult blood pressure categories are applied to children here for simplicity, so the youngest group should be read with care.

High blood pressure becomes more common with age in both genders

Men reach elevated SBP earlier, while women catch up after age 60



Source: NHANES, participants with average SBP

Figure 6: Share of participants in each systolic blood pressure category by age group, shown separately for females and males.